

## Assignment 1 Submission; due Thursday, September 24, 2026

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- [1] Let  $E \subseteq \mathbb{R}$  be nonempty. We say that  $t \in \mathbb{R}$  is a *lower bound* of  $E$  if  $t \leq x$  for all  $x \in E$ . We say that  $E$  is *bounded below* if there exists a lower bound of  $E$ . We say that  $t \in \mathbb{R}$  is a *greatest lower bound*, or *infimum*, of  $E$  if it is a lower bound of  $E$  and if  $t \geq s$  for any lower bound  $s$  of  $E$ . If  $E$  has an infimum, it is denoted  $\inf E$ .

- [a] Let  $E$  be a nonempty subset of  $\mathbb{R}$  which is bounded below. Find a way to use the completeness axiom to prove that  $\inf E$  exists. *Hint:* Consider the subset  $-E = \{-x : x \in E\}$ .

**Answer:**

*Proof.* Since  $E$  is nonempty and bounded below, there exists  $\alpha \in \mathbb{R}$  such that

$$\forall x \in E : \alpha \leq x$$

Consequently,

$$\forall x \in E : -\alpha \geq -x$$

.

Define

$$-E = \{-x : x \in E\}.$$

Then,  $-\alpha$  is an upper bound of  $-E$ .

By the completeness axiom,  $\sup(-E)$  exists. Let

$$\beta = \sup(-E).$$

It follows that  $\forall x \in -E : \beta \geq x$ . And therefore,  $\forall x \in E : -\beta \leq x$ . So  $-\beta$  is a lower bound of  $E$ .

Next, take any other lower bound of  $E$ , call it  $s$ . From the same logic as above, it follows that  $-s$  is an upper bound of  $-E$ . Since  $\beta$  is a supremum of  $-E$ , it follows that  $\beta \leq -s$ .

So,  $-\beta \geq s$ . Therefore, since  $-\beta$  is a lower bound of  $E$  and it is at least as large as any other lower bound, it follows that  $-\beta = \inf E$   $\square$

- [b] Let  $\emptyset \neq A \subseteq B \subseteq \mathbb{R}$ . Suppose  $B$  is bounded below (so  $\inf B$  exists). Prove that  $A$  is bounded below (so  $\inf A$  exists) and that  $\inf A \geq \inf B$ .

**Answer:**

*Proof.* If  $B$  is bounded below, then  $\exists \alpha : \forall x \in B : \alpha \leq x$ .

Since  $A \subseteq B$ , we know that  $\forall y \in A : y \in B$ .

Consequently,

$$\forall y \in A : \alpha \leq y$$

Therefore,  $\alpha$  is also a lower bound of  $A$ . And from 1[a], we can conclude that  $\inf A$  also exists.

Now, since  $A$  is a subset of  $B$  and  $\inf B$  is a lower bound of  $B$ , then  $\inf B$  must also be a lower bound of  $A$ . Since  $\inf A$  must be greater than or equal to any other lower bounds of  $A$ , we have that

$$\inf A \geq \inf B$$

□

- [c] Let  $A, B$  be nonempty subsets of  $\mathbb{R}$ , both bounded below (so both  $\inf A$  and  $\inf B$  exist). Define

$$A + B = \{a + b : a \in A, b \in B\}.$$

Prove that  $A + B$  is bounded below and that  $\inf(A + B) = \inf A + \inf B$ .

**Answer:**

*Proof.*

$$\begin{aligned} \forall a \in A : \inf A &\leq a \\ \forall b \in B : \inf B &\leq b \\ \therefore \forall a \in A, b \in B : \inf A + \inf B &\leq a + b \\ \therefore \forall x \in A + B : \inf A + \inf B &\leq x \end{aligned}$$

So  $\inf A + \inf B$  is a lower bound of  $A + B$ . Consequently,  $\inf(A + B)$  exists.

Now, we know that

$$\begin{aligned} \forall \varepsilon > 0 : \exists \alpha \in A : \inf A + \frac{\varepsilon}{2} &> \alpha \\ \forall \varepsilon > 0 : \exists \beta \in B : \inf B + \frac{\varepsilon}{2} &> \beta \end{aligned}$$

Therefore,

$$\forall \varepsilon > 0 : \exists \alpha \in A, \beta \in B : \inf A + \inf B + \varepsilon > \alpha + \beta$$

This means that no number greater than  $\inf A + \inf B$  can be a lower bound of  $A + B$ . Therefore,

$$\inf(A + B) = \inf A + \inf B$$

□

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- [2] Let  $x, y$  be two real numbers such that  $y - x > 1$ . (In particular,  $x < y$ .)

- [a] If  $x$  is positive, prove that there exists an integer  $k \in \mathbb{Z}$  such that  $x < k < y$ .

**Answer:**

*Proof.* By the Archimedean property, the set  $E = \{n \in \mathbb{N} : n > x\} \subset \mathbb{N} \subset \mathbb{Z}$  is not empty.

$x$  is a lower bound of  $E$ , so by the completeness axiom,  $\inf E$  exists. And since  $E \subset \mathbb{Z}$ ,  $\inf E \in E$ . Let  $\inf E = k$ .

Then  $x < k$ . Now, assume for the sake of contradiction that  $x < k - 1$ . Since  $k - 1 \in \mathbb{Z}$ , then  $k - 1 \in E$ . But  $k - 1 < k$  and  $k = \inf E$ , which is a contradiction.

Therefore,  $k - 1 \leq x$ .  $k \leq x + 1 < y$ .

So,  $x < k < y$ . □

- [b] If  $x$  is negative and  $y$  is positive, prove that there exists an integer  $k \in \mathbb{Z}$  such that  $x < k < y$ .

**Answer:**

*Proof.* Take  $k = 0 \in \mathbb{Z}$ . □

- [c] If  $y$  is negative, prove that there exists an integer  $k \in \mathbb{Z}$  such that  $x < k < y$ .

**Answer:**

*Proof.* Since  $y < 0$ , we have  $-y > 0$ . Moreover,

$$(-x) - (-y) = y - x > 1$$

So applying 2[a] to  $-x$  &  $-y$ , we get that  $\exists n \in \mathbb{Z} : -y < n < -x$ .

$$x < -n < y$$

Taking  $k = -n \in \mathbb{Z}$  proves the result. □

- [3] The parts of this problem are *not related*.

- [a] Let  $x \in \mathbb{R}$ . Consider the set  $E = \{q \in \mathbb{Q} : q < x\} \subset \mathbb{Q} \subset \mathbb{R}$ . Show that  $E$  is nonempty and bounded above, so  $\sup E$  exists. Show also that  $\sup E = x$ .

**Answer:**

*Proof.* Applying the Archimedean property to  $-x$  we get that  $\exists n \in \mathbb{N} : n > -x$ . So,  $x > -n$ . Therefore,  $-n \in E$ , so  $E$  is non-empty.

Also, since  $x$  is strictly larger than any element of  $E$ ,  $x$  is a lower bound of  $E$ .

Since  $E$  is bounded above and non-empty,  $\sup E$  exists.

Let  $\sup E = \alpha$ . We know that  $\alpha \leq x$ . Now, assume for the sake of contradiction, that  $\alpha < x$ .

One of the propositions proved in class showed that between any two distinct real numbers, there exists a rational number. That is:

$$\exists q \in \mathbb{Q} : \alpha < q < x$$

But if  $q$  is rational and smaller than  $x$ , then it should be a part of  $E$ . But this contradicts  $\alpha$  being an infimum of  $E$ . Therefore, our assumption must be false and

$$\sup E = x$$

□

- [b] Let  $x, y \in \mathbb{R}$  with  $x < y$ . Prove that there exists an *irrational* (not rational) number  $z$  such that  $x < z < y$ . *Hint:* You may use the fact that  $\sqrt{2}$  is irrational.

**Answer:**

*Proof.* We will take the interval  $(x, y)$  and shift it by  $\sqrt{2}$ .

$$x - \sqrt{2} < y - \sqrt{2}$$

By the proposition shown in class, we know that

$$\exists q \in \mathbb{Q} : x - \sqrt{2} < q < y - \sqrt{2}$$

Therefore,  $x < q + \sqrt{2} < y$ . Since we know that  $\sqrt{2}$  is irrational, then  $z = q + \sqrt{2}$  is also irrational. □

- [4] Let  $C > 0$  be a fixed positive number.

- [a] Let  $x \geq 0$ , and suppose that for any  $\varepsilon > 0$ , we have  $x < C\varepsilon$ . Show that for any  $\eta > 0$ , we have  $x < \eta$ .

**Answer:**

*Proof.* Let  $\eta > 0$  and choose

$$\varepsilon = \frac{\eta}{C}$$

Then we know that  $x < C\varepsilon = C \frac{\eta}{C}$ . Therefore,  $x < \eta$  □

- [b] In the situation of part [a], show that we must have  $x = 0$ . *Hint:* Find a way to use the Archimedean property.

**Answer:**

*Proof.* Suppose for the sake of contradiction that  $x > 0$ .

By the Archimedean property, we get that

$$\exists n \in \mathbb{N} : n > \frac{1}{x}$$

Therefore,  $x > \frac{1}{n}$ . But we can take  $\eta = \frac{1}{n} > 0$  in 4[a] and get that  $x < \frac{1}{n}$ . This is a contradiction, so our assumption must be wrong. Therefore,  $x = 0$ . □

- [5] In this problem you will give an alternative proof of the Cauchy–Schwarz inequality. The inequality says that

$$-\|x\|\|y\| \leq \langle x, y \rangle \leq \|x\|\|y\|.$$

Both inequalities are trivially seen to be equalities if either  $x = 0$  or  $y = 0$ , so we suppose that both vectors are nonzero.

By consider the norm of the vector

$$z = \frac{x}{\|x\|} \pm \frac{y}{\|y\|},$$

Show that:

1. both inequalities are true.
2. the first inequality becomes an equality if and only if  $y = tx$  for some  $t < 0$ .
3. the second inequality becomes an equality if and only if  $y = tx$  for some  $t > 0$ .

**Answer:**

*Proof.* Let  $u = \frac{x}{\|x\|}$ ,  $v = \frac{y}{\|y\|}$  and  $z = u \pm v$ .

$$\begin{aligned} \|u\| &= \left\| \frac{x}{\|x\|} \right\| = \frac{\|x\|}{\|x\|} = 1 \quad \|v\| = \left\| \frac{y}{\|y\|} \right\| = \frac{\|y\|}{\|y\|} = 1 \\ \|z\|^2 &= \|u \pm v\|^2 = \langle u \pm v, u \pm v \rangle = \|u\|^2 + \|v\|^2 \pm 2\langle u, v \rangle = 2 \pm 2\langle u, v \rangle \\ \|z\|^2 &= 2(1 \pm \langle u, v \rangle) \geq 0 \implies -1 \leq \langle u, v \rangle \leq 1 \\ \langle u, v \rangle &= \left\langle \frac{x}{\|x\|}, \frac{y}{\|y\|} \right\rangle = \frac{\langle x, y \rangle}{\|x\|\|y\|} \\ \therefore -1 &\leq \frac{\langle x, y \rangle}{\|x\|\|y\|} \leq 1 \implies -\|x\|\|y\| \leq \langle x, y \rangle \leq \|x\|\|y\| \end{aligned}$$

Next, for the equality conditions, we will first prove

$$y = tx \text{ for some } t > 0 \iff \langle x, y \rangle = \|x\|\|y\|$$

First, suppose  $y = tx$  for some  $t > 0$ . We have to prove that  $\langle x, y \rangle = \|x\|\|y\|$ .

$$\begin{aligned} \langle x, y \rangle &= \langle x, tx \rangle = t\langle x, x \rangle = t\|x\|^2 \\ \|y\| &= \|tx\| = t\|x\| \\ \therefore \langle x, y \rangle &= \|x\|\|y\| \end{aligned}$$

Secondly, suppose  $\langle x, y \rangle = \|x\|\|y\|$ . We have to prove that  $y = tx$  for some  $t > 0$ .

Since  $\langle x, y \rangle = \|x\|\|y\|$ , we have that  $\langle u, v \rangle = \frac{\langle x, y \rangle}{\|x\|\|y\|} = 1$ . Let  $z = u - v$ .

$$\begin{aligned} \|z\|^2 &= 2 - 2\langle u, v \rangle = 0 \implies u = v \\ \frac{x}{\|x\|} &= \frac{y}{\|y\|} \implies y = x \frac{\|y\|}{\|x\|} & \|y\| > 0, \quad \|x\| > 0 \\ \therefore y &= tx & \text{For } t = \frac{\|y\|}{\|x\|} > 0 \end{aligned}$$

This concludes the proof that  $y = tx$  for some  $t > 0 \iff \langle x, y \rangle = \|x\|\|y\|$

Now we have to prove

$$y = tx \text{ for some } t < 0 \iff \langle x, y \rangle = -\|x\|\|y\|$$

Consider the vector  $x' = -x$  and apply the proof above to show that

$$y = t'x' \text{ for some } t' > 0 \iff \langle x', y \rangle = \|x'\|\|y\|$$

Now,

$$y = t'(-x) \text{ for some } t' > 0 \iff \langle -x, y \rangle = \|-x\|\|y\|$$

$$y = (-t')x \text{ for some } t' > 0 \iff -\langle x, y \rangle = \|x\|\|y\| \quad \|x\| = \|-x\|$$

$$y = (-t')x \text{ for some } t' > 0 \iff \langle x, y \rangle = -\|x\|\|y\|$$

Take  $t = -t' < 0$

$$y = tx \text{ for some } t < 0 \iff \langle x, y \rangle = -\|x\|\|y\|$$

□

[6] Let

$$E = \overline{B_r(x_0)} = \{x \in \mathbb{R}^n : \|x - x_0\| \leq r\}$$

be a closed ball. Prove that  $E$  is *not open*.

**Answer:**

*Proof.* We have to prove that  $\exists x \in E : \forall \delta > 0 : B_\delta(x) \not\subseteq E$ .

Equivalently:  $\exists x \in E : \forall \delta > 0 : \exists y \notin E : \|y - x\| < \delta$ .

Take any  $x$  such that  $\|x - x_0\| = r$ . Then, for any  $\delta > 0$  construct the vector:

$$y = x + \frac{\delta}{2r}(x - x_0)$$

$$\|y - x_0\| = \left\| x + \frac{\delta}{2r}(x - x_0) - x_0 \right\| = \left\| (x - x_0)\left(1 + \frac{\delta}{2r}\right) \right\|$$

$$\|y - x_0\| = \left(1 + \frac{\delta}{2r}\right)\|x - x_0\| = \left(1 + \frac{\delta}{2r}\right)r = r + \frac{\delta}{2}$$

$$\|y - x_0\| = r + \frac{\delta}{2} > r$$

Therefore,  $y \notin E$ .

$$\|y - x\| = \left\| x + \frac{\delta}{2r}(x - x_0) - x \right\| = \left\| \frac{\delta}{2r}(x - x_0) \right\|$$

$$\|y - x\| = \frac{\delta\|x - x_0\|}{2r} = \frac{\delta r}{2r} = \frac{\delta}{2}$$

$$\therefore \|y - x\| = \frac{\delta}{2} < \delta$$

Thus, we have proved that  $\exists x \in E : \forall \delta > 0 : \exists y \notin E : \|y - x\| < \delta$ , which is equivalent to saying  $E$  is not open. □

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[7] Let  $F = \{x_1, \dots, x_N\}$  be a finite subset of  $\mathbb{R}^n$ . Prove that  $F$  is *closed*.

**Answer:**

*Proof.* We have to prove that  $\forall x \in F^c : \exists \delta > 0 : B_\delta(x) \subseteq F^c$ .

Let  $y \in F^c$ . So  $\forall x_i \in F : y \neq x_i$ . Therefore,  $\forall x_i \in F : \|y - x_i\| > 0$

Since  $F$  is a finite set, the set  $M = \{\|y - x_i\| : i \in [N]\}$  is also finite, so  $\min(M)$  exists.

Take  $\delta = \min(M) = \min(\|y - x_i\|) > 0$ .

Let  $z \in B_\delta(y) \implies \|z - y\| < \delta$ .

By the triangle inequality:  $\forall x_i \in F : \|z - x_i\| \geq \|y - x_i\| - \|y - z\|$ .

Now, since  $\delta = \min(\|y - x_i\|)$ , we know that  $\forall x_i \in F : \delta \leq \|y - x_i\|$

So, the inequality becomes:  $\forall x_i \in F : \|z - x_i\| \geq \delta - \|y - z\|$ .

But we know that  $\|y - z\| < \delta$ . Therefore,  $\forall x_i \in F : \|z - x_i\| > 0$ .

So,  $\forall x_i \in F : z \neq x_i \implies z \in F^c \implies B_\delta(y) \subseteq F^c$ .

Therefore  $F^c$  is open  $\implies F$  is closed. □

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[8] Let  $a \in \mathbb{R}^n \setminus \{0\}$ , and  $c \in \mathbb{R}$ . Consider the subset

$$E = \{x \in \mathbb{R}^n : \langle x, a \rangle < c\}.$$

Prove that  $E$  is open. *Hint:* Observe that

$$|\langle y, a \rangle - \langle z, a \rangle| = |\langle y - z, a \rangle| \leq \|y - z\| \|a\|$$

by Cauchy–Schwarz.

**Answer:**

*Proof.* Take arbitrary  $y \in E$ . Set

$$\delta = \frac{c - \langle y, a \rangle}{\|a\|}.$$

Since  $\langle y, a \rangle < c$  and  $\|a\| > 0$ , we have  $\delta > 0$ .

Let  $z \in B_\delta(y)$ . Then  $\|z - y\| < \delta$ , so by Cauchy–Schwarz,

$$\begin{aligned} \langle z, a \rangle &= \langle y, a \rangle + \langle z - y, a \rangle \\ &\leq \langle y, a \rangle + |\langle z - y, a \rangle| \\ &\leq \langle y, a \rangle + \|z - y\| \|a\| \\ &< \langle y, a \rangle + \delta \|a\| \\ &= c. \end{aligned}$$

Thus  $z \in E$ , and hence  $B_\delta(y) \subseteq E$ . Since  $y \in E$  was arbitrary,  $E$  is open. □

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- [9] Let  $C \subset \mathbb{R}^n$  be a non-empty proper *closed* subset of  $\mathbb{R}^n$ . Let  $x \in \mathbb{R}^n \setminus C$ , which exists because  $C$  is a proper subset. Define the subset  $E \subset \mathbb{R}$  by

$$E = \{r > 0 : B_r(x) \cap C = \emptyset\}.$$

- [a] Explain why  $E \neq \emptyset$

**Answer:**

- [b] Prove that there exists an upper bound for  $E$ .

**Answer:**

- [c] Thus, by the completeness axiom,  $\sup E$  exists.

Show that  $\sup E \leq \|x - y\|$  for all  $y \in C$

**Answer:**

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- [10] Let  $(x_k)_{k \in \mathbb{N}}$  be a sequence in  $\mathbb{R}^n$ , and suppose that there exists  $r \in (0, 1)$  and  $C > 0$  such that

$$\|x_{k+1} - x_k\| \leq Cr^k$$

Prove that  $(x_k)$  is Cauchy.

*Hint:* Recall that  $r^{k+m-1} + r^{k+m-2} + \dots + r^k = r^k(r^{m-1} + r^{m-2} + \dots + 1) = r^k \frac{1-r^m}{1-r}$ .

You may also use without proof the fact (which is easy to show using the Archimedean Property) that: since  $r \in (0, 1)$ , given any  $\delta > 0$ , there exists  $N \in \mathbb{N}$  such that  $k \geq N$  implies  $r^k < \delta$ .

**Answer:**